

Predicting Antimicrobial Resistance in Bacteria using a Specialized DNABERT2 and DBGWAS-Derived Features

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Abstract

Antimicrobial resistance (AMR) is an escalating public health threat, especially in hospitals where diverse resistance gene reservoirs have emerged. The COVID-19 pandemic intensified this issue through widespread antibiotic use in immunocompromised patients (Afshinnkoo et al., 2021). This highlights the urgent need for methods to rapidly detect AMR and uncover novel resistance mechanisms. With the increasing availability of metagenomic and whole-genome sequencing data from AMR pathogens, there is a timely opportunity to develop predictive models. Given the complexity of these genomic datasets, large language models (LLMs) offer a promising approach due to their ability to capture long-range sequence patterns. DNABERT2, an LLM pretrained on diverse DNA sequences, has shown strong performance in various genomic tasks and is well-suited for AMR prediction (Zhou et al., 2023). In this work, we present a novel method to predict AMR across nine pathogenic bacterial species treated with four common antibiotics. Four custom DNABERT2 models, pretrained on human microbiome-derived genomic sequences, were fine-tuned on sequences obtained from de novo assembled bacterial genomes. To extract phenotype-associated features, we employed De Bruijn Graph-based Genome-Wide Association Study (DBGWAS) in an alignment-free manner (Jaillard et al., 2018). Statistically significant sequences ($p < 0.05$) were aligned back to assemblies using BLAST ($\geq 80\%$ identity), and 1,000 bp flanking subsequences were extracted. Resistant samples showed a markedly higher number of BLAST hits than susceptible ones. Data were grouped by antibiotic (Gentamicin, Erythromycin, Ceftazidime, and Tetracycline) and species (9 different pathogens), and each group was fine-tuned using a DNABERT2 model incorporating species identity and BLAST hit count as additional features. Consensus predictions across sequences achieved 84.3% accuracy and a macro F1 score of 0.84. Using the original DNABERT2 model (without microbiome pretraining) resulted in only a 3% drop in accuracy, underscoring the generalizability of LLMs like DNABERT across diverse biological domains. Our findings demonstrate that resistant bacteria contain distinct genomic features absent in susceptible strains, highlighting the promise of LLM-based methods for AMR prediction.

Data Collection and Processing:

First, to create a model that specialized in understanding bacterial genomes, a DNABERT2 model was pretrained from scratch on sequences from the human microbiome. These sequences were obtained from publicly available databases from MGNify, including the Unified Human Gastrointestinal Genome (UHGG) v2.0.2 (Almeida et al., 2021), the human oral database, and the human vaginal database (Richardson et al., 2023). These datasets contain both WGS data of cultured bacterial isolates and metagenome assembled genomes (MAGs) from metagenomic shotgun sequencing of the human gut. The genomes were clustered based on similarity and grouped into 4,744 species-level clusters for the UHGG and 732 species-level clusters for the oral and vaginal databases. Each species-level cluster contained a

representative genome which was used and split into 1,000bp subsequences, sequences with greater than 80% similarity were removed using MMSeqs2 (Steinegger & Söding, 2017), and the resulting pretraining dataset contained over 1.8 million 1,000bp sequences.

To utilize the training data available from CAMDA 2025's AMR prediction challenge, the 5,458 unique SRA accessions were downloaded using sra-toolkit's fasterq-dump with the `--skip-technical` flag. This resulted in over 10,700 raw read fastq files from WGS of isolates from 9 different pathogenic bacterial species (*Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Neisseria gonorrhoeae*, *Streptococcus pneumoniae*, *Salmonella enterica*, *Staphylococcus aureus*, *Pseudomonas aeruginosa*, *Campylobacter jejuni*, and *Escherichia coli*) each treated with one of four different antibiotics (Gentamicin, Erythromycin, Ceftazidime, and Tetracycline), and containing the phenotypes Resistant (2,744 samples), Intermediate (684 samples), and Susceptible (2,716 samples). Intermediate phenotypes were treated as Resistant for training and testing. The raw reads then went through quality control and filtering using fastp (Chen et al., 2018) and de novo assembly using SPAdes (Bankevich et al., 2012) with the `--isolate` option to obtain longer contigs for further analysis without losing larger structural variant, indel, or novel resistance genes or plasmids which alignment to a reference genome would have failed to include. This resulted in on average 380 contigs per assembly with an average length of 24,000bp per contig, necessitating a way to extract only genomic regions relevant to AMR.

To accomplish this, we employed the De-Brujin Genome Wide Association Study (DBGWAS) program to extract sequences from the assemblies that were statistically associated with phenotype without the need for alignment to a reference genome. DBGWAS uses compact De-Brujin graphs to construct subgraphs representing phenotype-associated genetic variants and has been proven successful in extracting known resistance markers in antibiotic resistant bacteria (Jaillard et al., 2018). The program was run independently for each species with 2 phenotypes (treating Intermediate as Resistant) for *Campylobacter jejuni*, *Klebsiella pneumoniae*, *Neisseria gonorrhoeae*, and *Streptococcus pneumoniae*, and with 3 phenotypes (Resistant, Intermediate, and Susceptible) for the remaining species. This was due to the exceptionally long runtime for 2 phenotype analysis on the remaining species likely due to the high similarity of Intermediate to Susceptible phenotypes in these species. The program resulted in a list of variable length sequences and their associated p-values for each species. Sequences were filtered by p-value, using only sequences with a p-value of 0.05 or lower (multiple p-value thresholds were tested) for subsequent steps. These significant sequences were then aligned back to each of the original assemblies using BLAST with an 80% identity. The number of hits that each assembly's BLAST returned was taken and ultimately used as input to the classifier model due to the difference in distribution between Resistant and Susceptible phenotypes (figure 4). 1,000bp subsequences were then extracted from the assemblies by getting flanking regions around each BLAST hit. Assemblies that received less than 55 BLAST hits were topped up to 55 by extracting random 1,000bp subsequences from the assembly.

Methodology

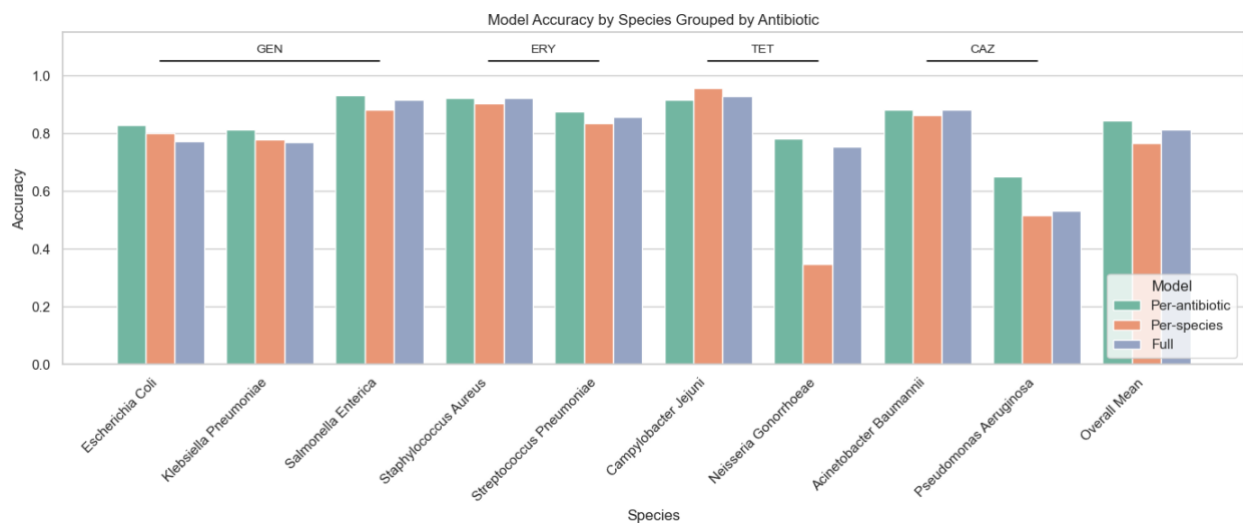
Three sets of finetuned DNABERT2 models were tested: (1) four separate models each trained on samples corresponding to a single antibiotic (taking species as an additional categorical variable) (per-antibiotic models), (2) nine separate models each trained on samples corresponding to a single species (taking antibiotic as an additional categorical variable) (per-species models), and (3) one single model trained on data from all samples together, with antibiotic and species as additional features (subsampled down to 500 rows due to memory constraints) (full model). Due to a class imbalance of approximately 2 times the number of Resistant phenotype sequences as Susceptible, the training set had the Susceptible rows augmented 2 times by computing the reverse complement of their sequence and keeping all other features the same. The testing and validation sets were left as is to mimic the final testing dataset.

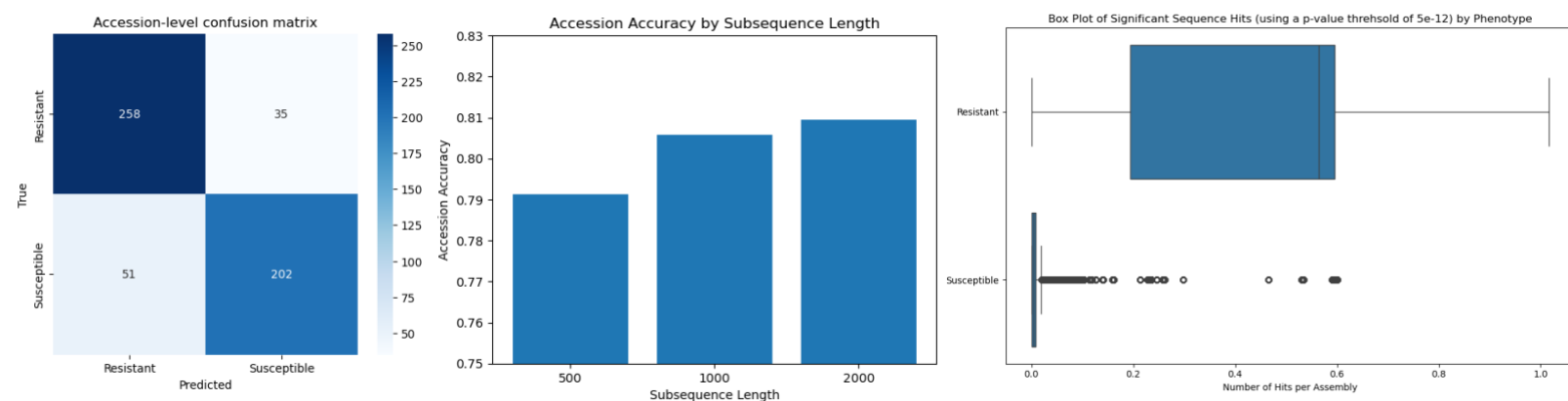
During training, a Low-Rank Adaption (LoRA) technique was used to finetune the pretrained bacterial model on the sequences in the train set. After each forward pass through the pretrained model, a 10% dropout rate was applied to the pooled embeddings, and they were concatenated with the 8 dimensional species and/or antibiotic embeddings and normalized number of BLAST hits. The full-feature embedding was then passed through a 3-layer classifier with a hidden layer size of 128 and a ReLU activation function. Loss on the binary labels was calculated using a cross-entropy loss function.

The test set went through the same pipeline except for the data augmentation and random sequence extraction step. Only sequences with no BLAST hits had a single random subsequence extracted. To make final predictions, the test set was grouped by accession number and a consensus was taken based on the most frequent prediction by the model for each sample. Each model was evaluated on accession-level accuracy, accession-level f1 score, and sequence-level accuracy.

Results

The number of BLAST hits of DBGWAS-derived sequences with a p-value lower than $5e-12$ showed an elevated distribution for Resistant samples than for Susceptible (figure 4). Using a DBGWAS p-value threshold of 0.05, a BLAST percent identity of 80%, and a subsequence length of 1,000bp gave the best results. The per-antibiotic models gave the best overall accuracy of 84.5% and for each species individually except for one (figure 1) on the testing subset of the training data. The evaluation on the CAMDA provided test set gave an accuracy overall accuracy of 82.8%, f1 score of .829, and aggregate phenotype balanced accuracy of 80.2%. The confusion matrix revealed a higher proportion of Susceptible samples predicted as Resistant than the reverse (figure 2). The Erythromycin model gave the best performance with an accession-level accuracy of 90% and the Ceftazidime model gave the worst with an accession-level accuracy of 77% (figure 3). Using input lengths of 500, 1000, and 2000bp was also tested and showed that there was a minimal performance increase from 1,000 to 2,000bp (figure 3). The original pretrained DNABERT2 model was also evaluated instead of the bacterial pretrained model and performed with about 3% lower accession-level accuracy.





All figures refer to consensus prediction results on the 10% testing subset of the CAMDA provided training data (1) Accuracy of each model by species grouped by antibiotic (2) Overall confusion matrix on binary classification (3) Full model performance by 500, 1000, and 2000bp training and testing length (4) Number of BLAST hits of DBGWAS-derived sequences with p-values less than 5e-12 by phenotype

Discussion

The difference in distributions of BLAST hits across Resistant and Susceptible phenotypes (figure 1) shows that DBGWAS is most likely extracting sequences from genes and plasmids that enable AMR, which are present in Resistant assemblies but not in Susceptible ones. The remaining sequences that did BLAST to Susceptible phenotypes could be associated with genes or regulatory elements containing variants that enable AMR. The superior performance of the per-antibiotic models compared to the full and per-species models suggests that AMR genes and plasmids are not species-specific, but rather can be shared across different pathogenic species through mechanisms such as transformation and horizontal gene transfer-- a phenomenon that has been previously confirmed (Barcan et al., 2025). This finding is particularly concerning in environments where symbiotic bacteria acquire resistance due to antibiotic overuse, potentially passing these traits to pathogenic species.

Although the domain-specific model performed slightly better than the original pretrained model, the minor loss in performance highlights the ability of these models to generalize even across different domains of life. The performance of these models demonstrates the ability of DBGWAS to extract discriminative genomic regions for AMR. Future work should investigate these regions and determine their origin (horizontal gene transfer, plasmids, or spontaneous mutations) and mechanism of resistance. This work could enable specific guide RNAs to be engineered to target and disrupt these regions using CRISPR/Cas9, disabling AMR in pathogenic bacteria.

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